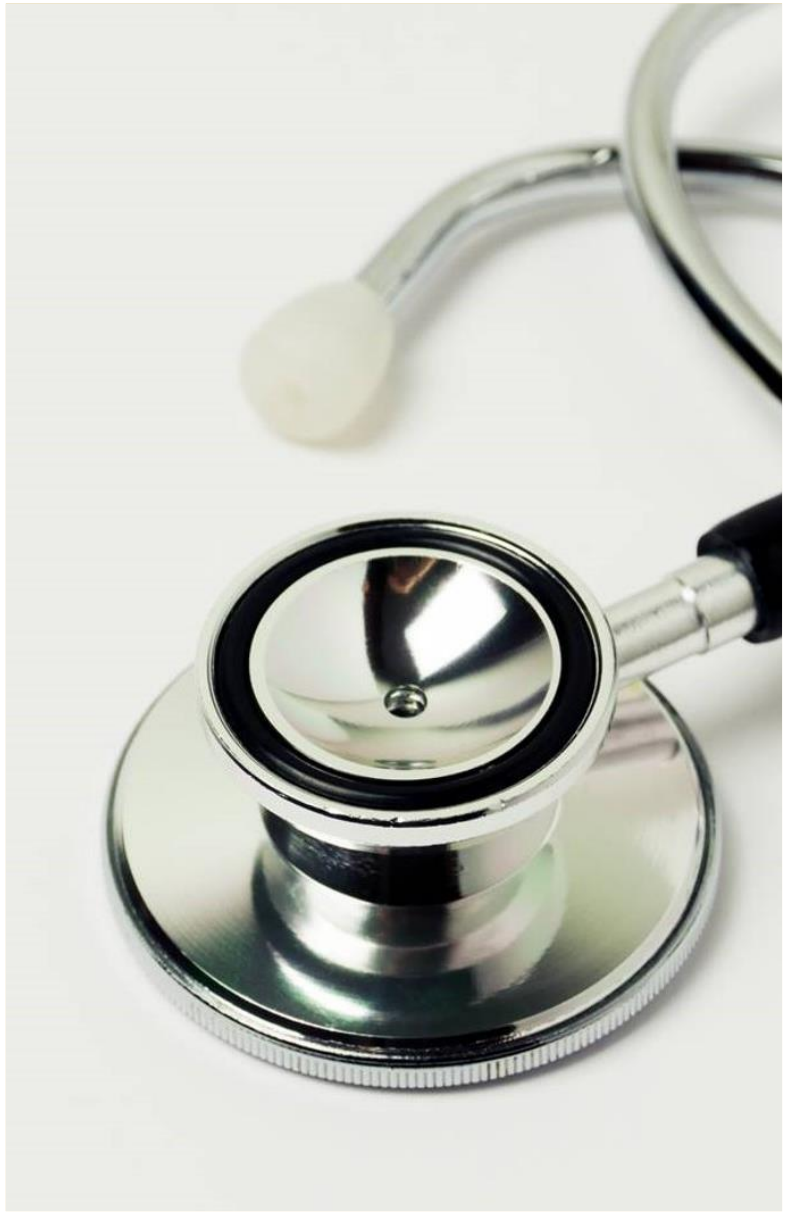
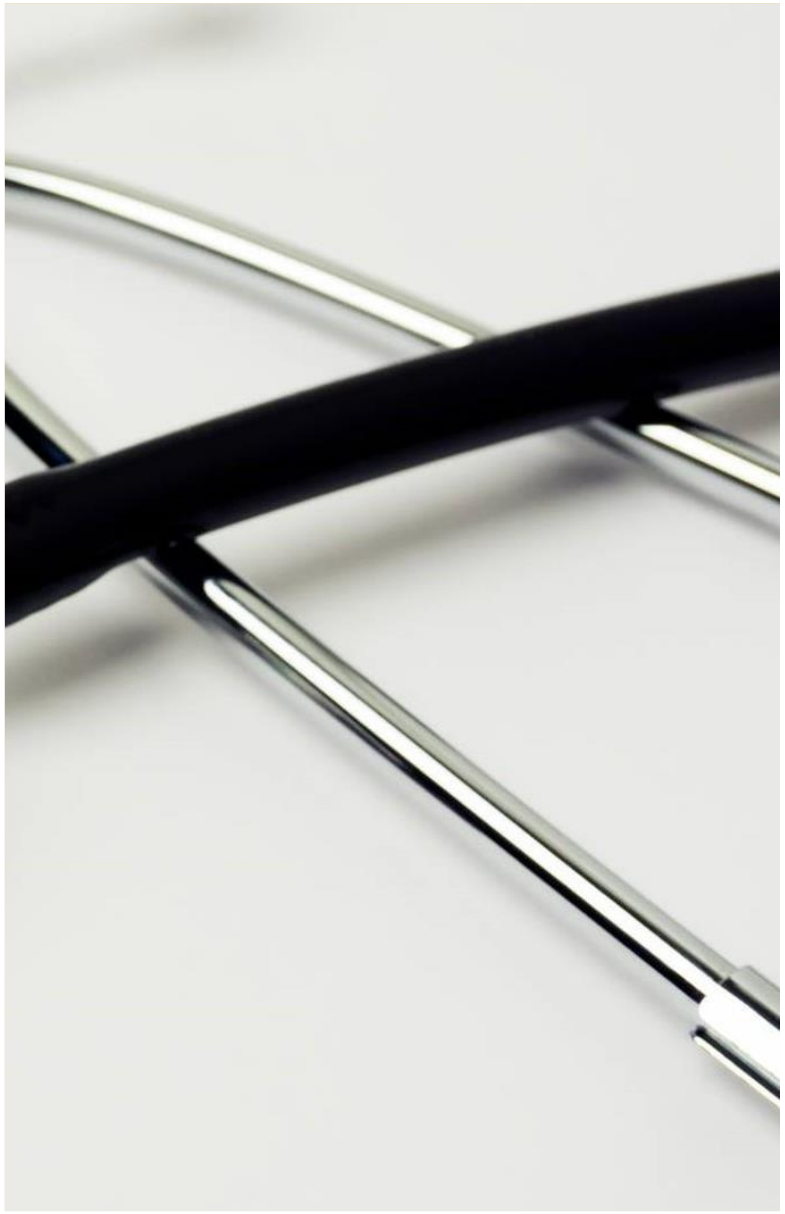
DISORDERS OF CARDIAC FUNCTION, HEART FAILURE AND CIRCULATORY SHOCK



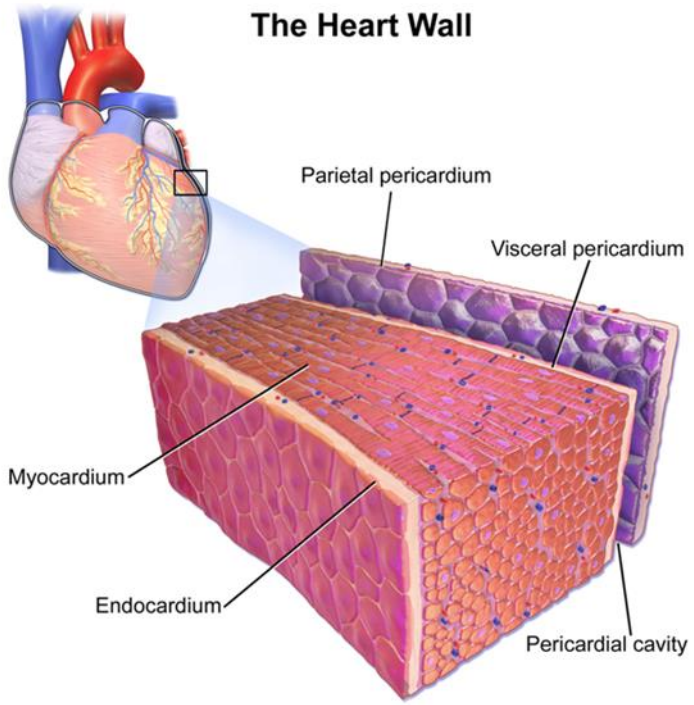
AGENDA

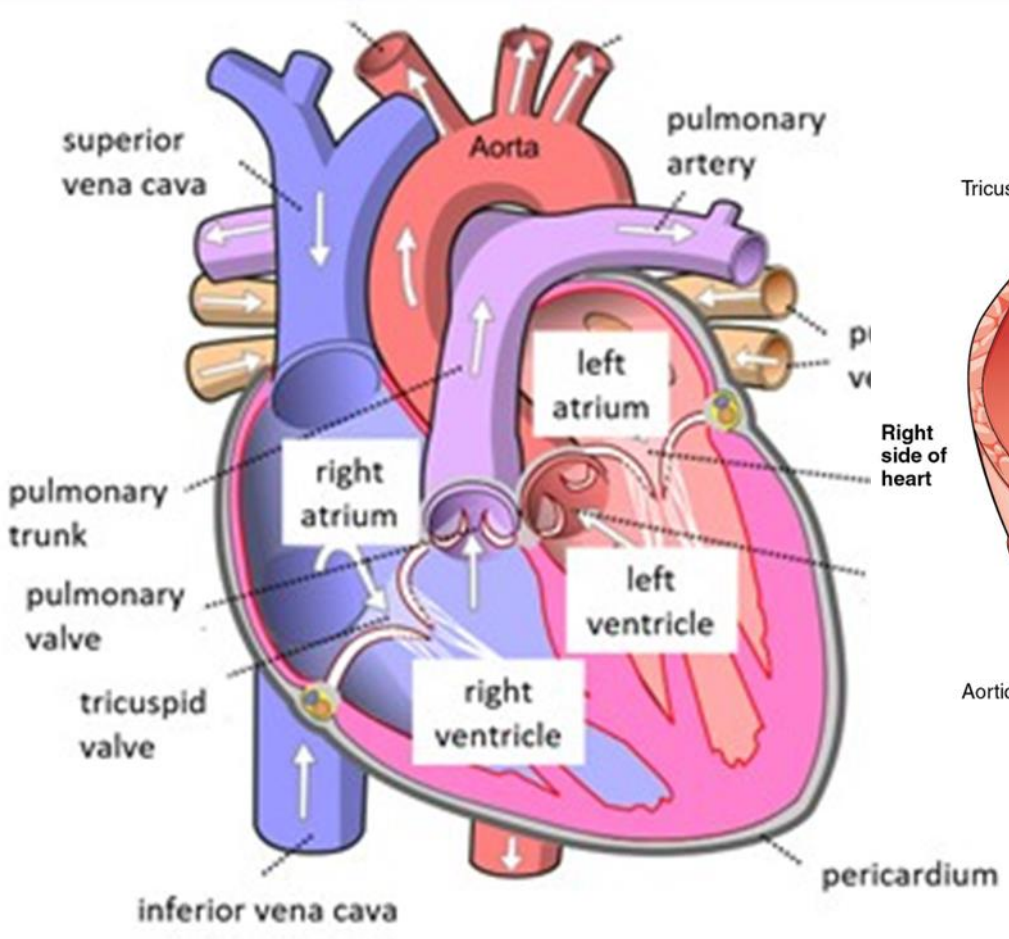
1. Explain the pathophysiology of key cardiovascular disorders.
2. Identify risk factors, clinical manifestations, and complications.
3. Discuss the treatment and management approaches.

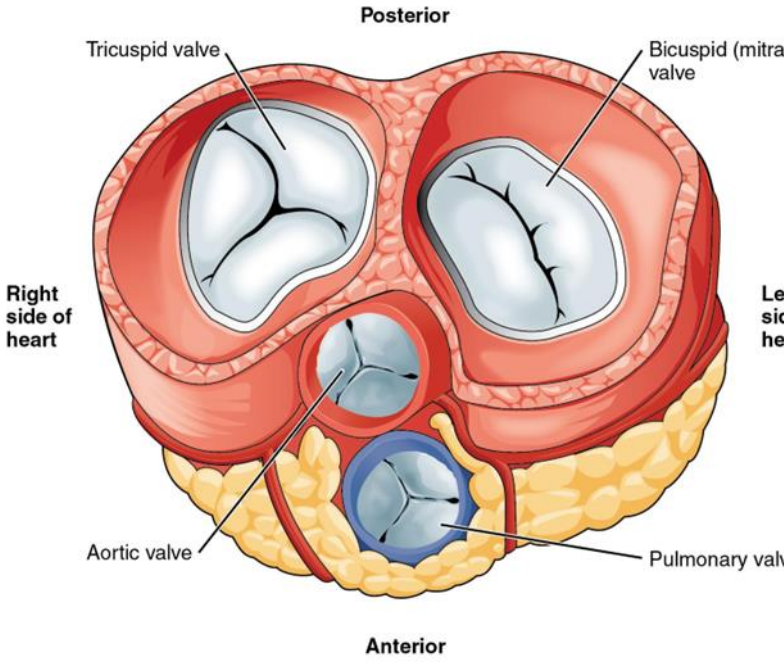




OVERVIEW OF CARDIOVASCULAR SYSTEM DISORDERS







PERICARDIAL DISORDERS

- **Definition:** Inflammation or infection of the pericardium (membrane surrounding the heart).
- **Key Disorders:**
  - **Pericarditis**: Inflammation leading to chest pain, fluid accumulation.
  - **Pericardial Effusion**: Excess fluid accumulation in the pericardial sac.
  - **Cardiac Tamponade**: Pressure on the heart due to fluid accumulation, impairing heart function.
- **Pathophysiology:** Inflammation from infections, autoimmune disorders, or post-surgical. Fluid buildup compresses the heart, reducing cardiac output.
- **Clinical Manifestations:** Sharp chest pain, dyspnea, hypotension, tachycardia, muffled heart sounds.

CORONARY HEART DISEASE (CHD)

- **Definition:**
  Coronary heart disease (CHD), also called coronary artery disease (CAD), occurs when the coronary arteries become narrowed or blocked due to atherosclerosis, leading to reduced blood flow to the heart muscle.
- **Pathophysiology:**
- **Atherosclerosis**: The primary cause of CHD is the buildup of plaque (fatty deposits) inside the coronary arteries. This plaque consists of cholesterol, fatty substances, and cellular waste products.

MYOCARDITIS

- **Definition:** Inflammation of the heart muscle, often caused by viral infections.
- **Pathophysiology:** Viral infection (e.g., Coxsackievirus) leads to inflammatory response in the myocardium, resulting in myocardial dysfunction.
- **Clinical Manifestations:** Fever, fatigue, chest pain, arrhythmias, heart failure signs.

CORONARY HEART DISEASE (CHD)

- Over time, plaque can harden and narrow the arteries, restricting blood flow and oxygen to the heart muscle, which can lead to myocardial ischemia (insufficient blood flow).
- Plaque rupture: If a plaque ruptures, it can cause a blood clot to form, which further obstructs the coronary artery and can lead to a myocardial infarction (MI) (heart attack).
- Ischemia: Lack of oxygen in the heart muscle leads to chest pain (angina) and may result in irreversible damage to the heart tissue if blood flow is not restored.

TYPES OF CHD:

- **Stable angina**: Chest pain that occurs predictably with exertion or stress and is relieved by rest or nitroglycerin.
- **Unstable angina**: Chest pain that occurs unpredictably or at rest and is more intense, signifying a high risk of myocardial infarction.
- **Myocardial infarction (MI)**: A heart attack caused by the complete blockage of a coronary artery, leading to the death of heart muscle cells.

RISK FACTORS, & CLINICAL MANIFESTATIONS

- Hypertension
- Hyperlipidemia (high cholesterol)
- Smoking
- Diabetes mellitus
- Obesity
- Sedentary lifestyle
- Family history of cardiovascular diseases

- **Angina** (chest pain)
- Shortness of breath
- Fatigue
- Sweating
- Nausea
- **Myocardial infarction (MI)** symptoms: Severe chest pain, pain radiating to the arm, jaw, back, or stomach, and sweating.

MANAGEMENT:

- **Medications**: Antiplatelets (e.g., aspirin), statins (cholesterol-lowering), beta-blockers, nitroglycerin, ACE inhibitors.
- **Procedures**: Angioplasty, stent placement, coronary artery bypass grafting (CABG)

HEART FAILURE

- **Definition:**
  Heart failure (HF) is a condition in which the heart is unable to pump blood efficiently to meet the body's needs, leading to congestion in the lungs and/or systemic circulation.
- **Pathophysiology:**

- **Systolic heart failure**: Inadequate contraction of the heart muscle, leading to a reduced ejection fraction (EF) < 40%.
- **Diastolic heart failure**: Impaired relaxation and filling of the heart, with a normal or near-normal ejection fraction.

Pathophysiology cont..:

- **Compensatory mechanisms**: The body compensates for reduced cardiac output by activating the sympathetic nervous system (increasing heart rate), renin-angiotensin-aldosterone system (RAAS) (increasing blood volume), and releasing antidiuretic hormone (ADH) to retain water and sodium. While initially helpful, over time, these mechanisms worsen heart failure and lead to fluid retention, edema, and increased workload on the heart.

TYPES OF HEART FAILURE:

- **Left-sided heart failure**: The left ventricle is unable to pump blood effectively, leading to pulmonary congestion (fluid in the lungs).
  - **Clinical Manifestations**: Shortness of breath, orthopnea (difficulty breathing when lying flat), paroxysmal nocturnal dyspnea (waking up breathless), crackles in the lungs, fatigue.
- **Right-sided heart failure**: The right ventricle is unable to pump blood effectively into the lungs, leading to systemic fluid retention.
  - **Clinical Manifestations**: Peripheral edema (swelling of legs, ankles), jugular venous distension, hepatomegaly (enlarged liver), ascites (fluid in the abdomen).
- **Congestive heart failure**: A combination of both left and right-sided heart failure, with widespread fluid retention.

RISK FACTORS & CLINICAL MANIFESTATIONS:

- Hypertension
- Coronary artery disease (CHD)
- Myocardial infarction (heart attack)
- Valvular heart disease
- Diabetes
- Obesity

- Dyspnea (difficulty breathing)
- Fatigue
- Edema (swelling)
- Orthopnea (difficulty breathing while lying down)
- Weight gain (due to fluid retention)
- Tachycardia
- Decreased exercise tolerance

MANAGEMENT:

- **Pharmacological Treatment**: Diuretics (to reduce fluid overload), ACE inhibitors, beta-blockers, vasodilators, aldosterone antagonists.
- **Devices**: Implantable cardioverter-defibrillators (ICD), biventricular pacemaker (cardiac resynchronization therapy).
- **Surgical Intervention**: Heart transplant in severe case

CONGENITAL HEART DISEASE (CHD)

- **Definition:**
  Congenital heart disease refers to structural defects of the heart present at birth, affecting the heart's normal function.
- **Types of Congenital Heart Disease:**
- **Atrial Septal Defect (ASD)**: A hole between the two atria, allowing oxygen-rich blood to flow from the left atrium to the right atrium, increasing blood flow to the lungs.
  - **Clinical Manifestations**: Fatigue, shortness of breath, frequent respiratory infections, heart murmur.

CONGENITAL HEART DISEASE (CHD)

- Ventricular Septal Defect (VSD): A hole between the two ventricles, causing oxygen-rich blood from the left ventricle to mix with oxygen-poor blood from the right ventricle.
  - Clinical Manifestations: Cyanosis, fatigue, growth retardation, heart murmur.
- Patent Ductus Arteriosus (PDA): The ductus arteriosus fails to close after birth, causing blood to flow abnormally between the aorta and pulmonary artery.
  - Clinical Manifestations: Murmur, rapid breathing, fatigue, poor feeding

CONGENITAL HEART DISEASE (CHD)

- **Pathophysiology:**
- Abnormal heart development during fetal life results in improper formation of heart chambers, valves, or blood vessels, causing either abnormal blood flow or mixing of oxygenated and deoxygenated blood.
- **Management:**
- **Surgical Repair**: Most congenital heart defects require surgical intervention, such as patching holes or repairing abnormal connections.
- **Medications**: Diuretics, ACE inhibitors for congestive symptom

SHOCK

- **Definition:**
  Shock is a life-threatening condition where there is insufficient blood flow to tissues, resulting in inadequate oxygen and nutrient delivery, and cellular dysfunction.
- **Types of Shock:**
- **Hypovolemic Shock**: Caused by a significant loss of blood or fluids, reducing the circulatory volume and leading to reduced cardiac output.
  - **Causes**: Hemorrhage, dehydration, severe burns, vomiting, or diarrhea.
  - **Clinical Manifestations**: Tachycardia, hypotension, pale and cool skin, weak pulse, low urine output, confusion

Types of Shock

- **Cardiogenic Shock**: Caused by failure of the heart to pump blood effectively, often due to myocardial infarction or severe heart failure.
  - **Clinical Manifestations**: Chest pain, hypotension, tachycardia, cyanosis, confusion, organ failure.
- **Septic Shock**: Caused by severe infection, which leads to vasodilation, decreased blood pressure, and poor tissue perfusion.
  - **Clinical Manifestations**: Fever, hypotension, tachycardia, organ dysfunction, warm skin (early stages), cold and mottled skin (late stages)
- **Neurogenic Shock**: Caused by a sudden loss of sympathetic nervous system activity, leading to vasodilation and decreased systemic vascular resistance.
- **Causes**: Spinal cord injury, anesthesia, or drug overdose.
- **Clinical Manifestations**: Hypotension, bradycardia, warm skin, and respiratory distress

PATHOPHYSIOLOGY OF SHOCK:

- **Inadequate perfusion** leads to tissue hypoxia, anaerobic metabolism, and accumulation of lactic acid, causing metabolic acidosis.
- Compensatory mechanisms like tachycardia, vasoconstriction, and increased respiration attempt to maintain blood pressure and perfusion, but if untreated, shock progresses to organ failure and death.

CLINICAL MANIFESTATIONS & MANAGEMENT

- Tachycardia
- Hypotension
- Cool, clammy skin (except in septic shock)
- Rapid, shallow breathing
- Decreased urine output
- Altered mental status

- **Fluid resuscitation**: IV fluids to restore blood volume.
- **Medications**: Vasopressors (e.g., norepinephrine, dopamine), inotropes (e.g., dobutamine).
- **Oxygen therapy**: To support tissue oxygenation.
- **Treat underlying cause**: Infection, bleeding, or heart failure

THANK YOU

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